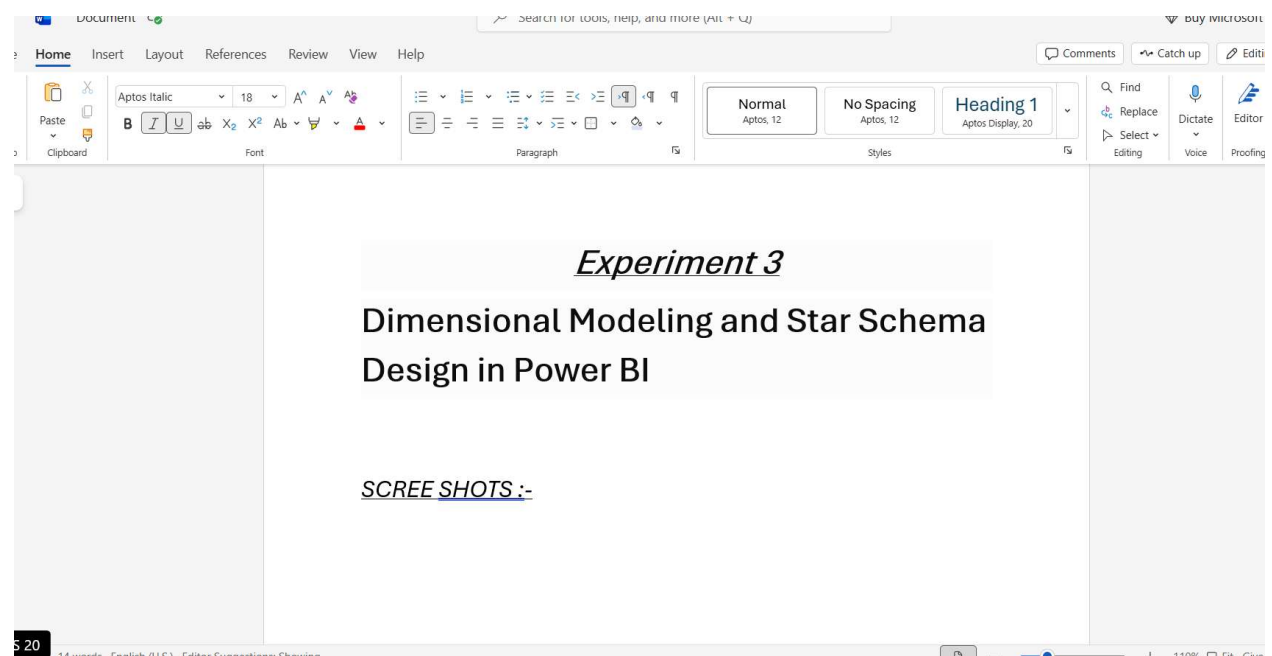
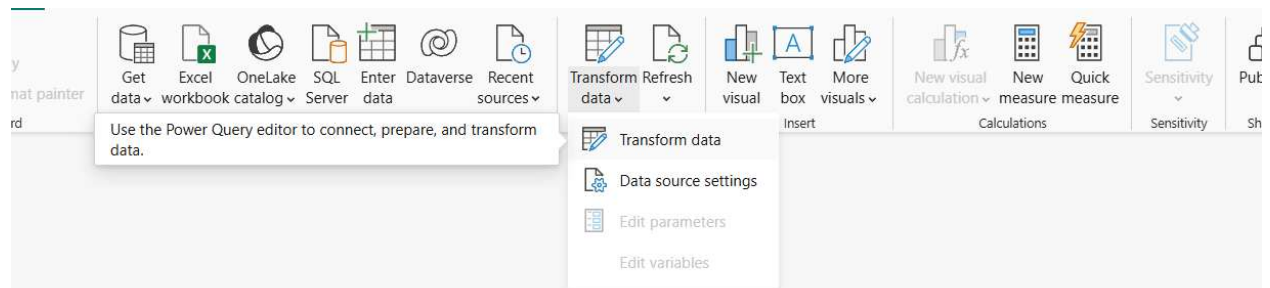


Experiment 3

Dimensional Modeling and Star Schema Design in Power BI

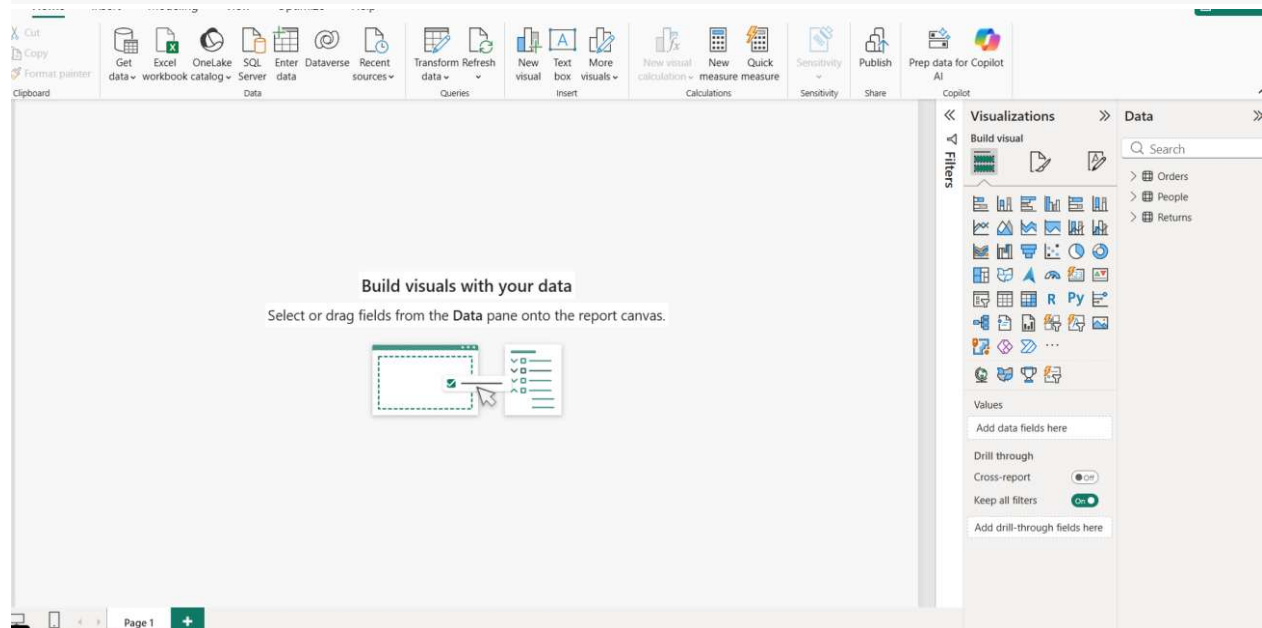
SCREE SHOTS :-





Build visuals with your data

Select or drag fields from the Data pane onto the report canvas.



Insert Modeling View Optimize Help

Get data Excel OneLake workbook catalog

Navigator

Display Options

- Global Superstore 2018 (1).xlsx [3]
 - Orders
 - People
 - Returns

Returns

Column1	Column2	Column3
Returned	Order ID	Region
Yes	IN-2017-CA120551-42816	Southern Asia
Yes	IN-2017-AA103751-42926	Southern Asia
Yes	IN-2017-TS212051-42904	Southern Asia
Yes	AG-2014-RO97803-41695	North Africa
Yes	AG-2015-LC70503-42265	North Africa
Yes	AG-2014-DB32103-41987	North Africa
Yes	AG-2017-AC4203-42796	North Africa
Yes	AG-2017-RD96603-43073	North Africa
Yes	AG-2015-CP23403-42152	North Africa
Yes	AG-2016-HF49953-42437	North Africa
Yes	AG-2015-CC24303-42247	North Africa
Yes	AG-2017-MO78003-42948	North Africa
Yes	AO-2017-BE14554-42954	Central Africa
Yes	AO-2017-GM45004-42996	Central Africa
Yes	AO-2016-IG50854-42668	Central Africa
Yes	US-2017-SR204255-42937	South America
Yes	US-2014-JP155205-41883	South America
Yes	MX-2016-LC171405-42457	South America
Yes	US-2016-JK152055-42585	South America
Yes	US-2014-JC153405-41897	South America
Yes	US-2017-CA119655-43061	South America
Yes	US-2016-AJ109455-42595	South America
Yes	US-2016-NB185805-42648	South America

Load Transform Data Cancel

Downloads

Search Downloads

Organize New folder

NAMAN - Person

Name	Date modified	Type	Size
Global Superstore 2018 (3)	21-08-2026 19:32	Microsoft Excel W...	7,1
Employee_DB (2)	21-08-2026 18:54	Microsoft Excel W...	7,1
Global Superstore 2018 (2)	17-08-2026 11:47	Microsoft Excel W...	7,1
Global Superstore 2018 (1)	17-08-2026 11:47	Microsoft Excel W...	7,1
Employee_DB (1)	17-08-2026 11:46	Microsoft Excel W...	7,1
Employee_DB	17-08-2026 11:38	Microsoft Excel W...	7,1
Global Superstore 2018	03-08-2026 11:35	Microsoft Excel W...	7,1

File name: Global Superstore 2018 (2)

Excel Files (*.xls;*.xlsx;*.xlsm;*.xlt)

Open Cancel

Import data from Excel Import data from SQL Server Paste data into a blank table Use sample data

Get data from another source →

Visualizations

Build visual

Filters

Values

Add data fields here

Drill through

Cross-report

Keep all filters

Add drill-through fields here

Power Query Editor interface showing a table with columns: Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, and Customer Name. The table contains 16 rows of data. The interface includes a ribbon with various options like Close & Apply, New, Recent, Enter Data, Data source settings, Manage Parameters, Export query results, Refresh Preview, Properties, Advanced Editor, Choose Columns, Remove Columns, Keep Rows, Reduce Rows, Sort, Split Columns, Group By, Transform, Data Type: Text, Use First Row as Headers, Replace Values, Merge Queries, Append Queries, and Combine Files.

Query Settings: Name: Orders, All Properties, APPLIED STEPS: Source, Navigation, Promoted Headers, Changed Type.

Power Query Editor interface showing a table with columns: City, State, Country, and Region. The table contains 7 rows of data. The interface includes a ribbon with various options like Close & Apply, New, Recent, Enter Data, Data source settings, Manage Parameters, Export query results, Refresh Preview, Properties, Advanced Editor, Choose Columns, Remove Columns, Keep Rows, Reduce Rows, Sort, Split Columns, Group By, Transform, Data Type: Text, Use First Row as Headers, Replace Values, Merge Queries, Append Queries, and Combine Files.

Column statistics: Count: 23, Error: 0, Empty: 0, Distinct: 23, Unique: 23, Empty string: 0, Min: Canada, Max: Western... Value distribution: Southern Asia, Southern Europe, North Africa, Central Africa, South America, Western Asia, Oceania, Western Europe, Caribbean, Eastern Europe, Central America.

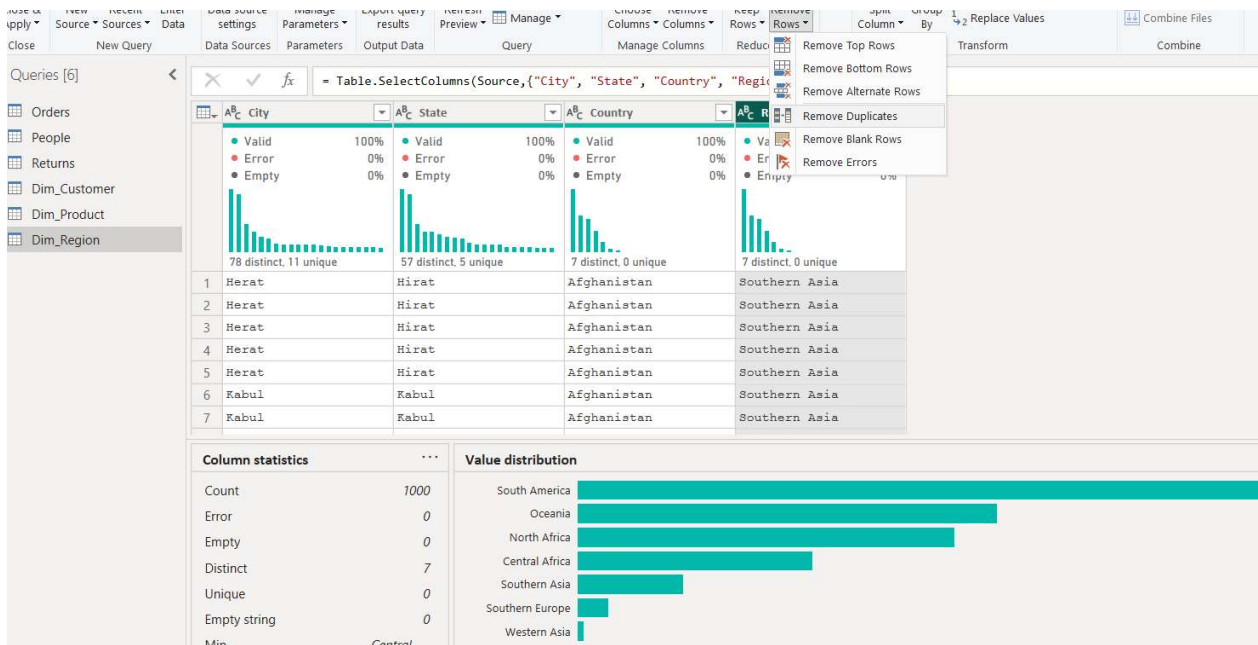


Table.SelectColumns(Source, {"Product ID", "Product Name", "Sub-Category"})

Product ID	Product Name	Sub-Category
16	FUR-TA-3758	Chromcraft Coffee Table, with Bottom Storage
17	TEC-PH-5251	Motorola Headset, Cordless
18	OFF-ST-5708	Rogers Shelving, Wire Frame
19	FUR-FU-5727	Rubbermaid Frame, Black
20	TEC-PH-5812	Samsung Audio Dock, Cordless
21	OFF-PA-5890	SanDisk Parchment Paper, 8.5 x 11

Refresh Preview

Advanced Editor

Manage

Query

Order ID

lid 100% Valid
ror 0% Error
tpty 0% Empty

Distinct, 242 unique

394 distinct

017-CA120551-428...

015-BD116051-422...

017-CA120551-428...

017-CA120551-428...

015-BD116051-422...

015-AJ107801-421...

017-GM144551-429...

017-VB217451-430...

016-LO171701-426...

017-VB217451-430...

015-SS201401-423...

017-AA103751-429...

015-BG110351-422...

016-AH105851-427...

016-CS118451-423...

016-CS118451-423...

015-BG110351-422...

017-AA103751-429...

014-AH105851-419...

015-DW131951-421...

014-RS194201-418...

Choose Columns

Choose the columns to keep

Search Columns

☒ (Select All Columns)

☐ Row ID

☐ Order ID

☐ Order Date

☐ Ship Date

☐ Ship Mode

☐ Customer ID

☐ Customer Name

☐ Segment

☐ Postal Code

☐ City

☐ State

☐ Country

☐ Region

☐ Market

☒ Product ID

☒ Product Name

☒ Sub-Category

☒ Category

☐ Sales

OK

Cancel

30-11-2014

03-12-2014

First Class

05-06-2015

10-06-2015

Standard Class

16-08-2014

18-08-2014

First Class

Close & Apply | New | Recent | Enter Data | Data source settings | Manage Parameters | Export query results | Refresh Preview | Properties | Choose Columns | Remove Columns | Keep Rows | Remove Rows | Sort | Split Column | Group By | Data Type: Whole Number | Use First Row as Headers | Replace Values | Merge Queries | Append Queries | Combine Files | Combine

Queries [5] | Orders | People | Returns | Dim_Customer | Dim_Product

Order ID | Order Date | Ship Date | Ship Mode | Customer ID | Customer Name

1000 distinct, 1000 unique | 494 distinct, 242 unique | 394 distinct, 152 unique | 404 distinct, 155 unique | 4 distinct, 0 unique | 449 distinct, 199 unique | 362 distinct, 122

1	24599	IN-2017-CA120551-428...	22-03-2017	29-03-2017	Standard Class	CA-120551	Cathy Arm
2	29465	ID-2015-BD116051-422...	01-09-2015	04-09-2015	Second Class	BD-116051	Brian Dah
3	24598	IN-2017-CA120551-428...	22-03-2017	29-03-2017	Standard Class	CA-120551	Cathy Arm
4	24597	IN-2017-CA120551-428...	22-03-2017	29-03-2017	Standard Class	CA-120551	Cathy Arm
5	29464	ID-2015-BD116051-422...	01-09-2015	04-09-2015	Second Class	BD-116051	Brian Dah
6	28879	ID-2015-AJ107801-421...	19-04-2015	22-04-2015	First Class	AJ-107801	Anthony J
7	27993	IN-2017-GM144551-429...	01-08-2017	05-08-2017	Standard Class	GM-144551	Gary Mitc
8	28967	IN-2017-VB217451-430...	11-12-2017	15-12-2017	Standard Class	VB-217451	Victoria
9	29492	IN-2016-LO171701-426...	24-09-2016	28-09-2016	Standard Class	LO-171701	Lori Olao
10	28966	IN-2017-VB217451-430...	11-12-2017	15-12-2017	Standard Class	VB-217451	Victoria
11	25232	ID-2015-SS201401-423...	16-12-2015	20-12-2015	Standard Class	SS-201401	Saphhira
12	23222	IN-2017-AA103751-429...	10-07-2017	15-07-2017	Second Class	AA-103751	Allen Arm
13	29094	IN-2015-BG110351-422...	28-09-2015	04-10-2015	Standard Class	BG-110351	Barry Gon
14	28265	IN-2016-AH105851-427...	27-11-2016	01-12-2016	Standard Class	AH-105851	Angele Ho
15	27278	IN-2016-CS118451-423...	18-01-2016	20-01-2016	First Class	CS-118451	Cari Sayr
16	27279	IN-2016-CS118451-423...	18-01-2016	20-01-2016	First Class	CS-118451	Cari Sayr
17	29096	IN-2015-BG110351-422...	28-09-2015	04-10-2015	Standard Class	BG-110351	Barry Gon

PROPERTIES | Dim_Product | All Properties | APPLIED STEPS | Source

Queries [5] | Orders | People | Returns | Dim_Customer | Orders (2)

Copy | Paste | Delete | Rename | Enable load | Include in report refresh | Duplicate | Reference | Move To Group | Move Up | Move Down | Create Function... | Convert To Parameter | Advanced Editor | Properties...

Order ID | Order Date | Ship Date | Ship Mode | Customer ID | Customer Name

1000 distinct, 1000 unique | 494 distinct, 242 unique | 394 distinct, 152 unique | 404 distinct, 155 unique | 4 distinct, 0 unique | 449 distinct, 199 unique | 362 distinct, 122

15	27278	IN-2016-CS118451-423...	18-01-2016	20-01-2016	First Class	CS-118451	Cari Sayr
16	27279	IN-2016-CS118451-423...	18-01-2016	20-01-2016	First Class	CS-118451	Cari Sayr
17	29096	IN-2015-BG110351-422...	28-09-2015	04-10-2015	Standard Class	BG-110351	Barry Gon
18	23219	IN-2017-AA103751-429...	10-07-2017	15-07-2017	Second Class	AA-103751	Allen Arm
19	28276	IN-2014-AH105851-419...	30-11-2014	03-12-2014	First Class	AH-105851	Angele Ho
20	29585	IN-2015-DW131951-421...	05-06-2015	10-06-2015	Standard Class	DW-131951	David Wie
21	23951	IN-2014-RS194201-418...	16-08-2014	18-08-2014	First Class	RS-194201	Ricardo S

The screenshot displays the Qlik Sense interface with a data table and several panels. The table has columns for Order ID, Order Date, Ship Date, Ship Mode, Customer ID, and Customer Name. The data is filtered by 'Valid' status. A context menu is open over the first row, showing options like Copy, Paste, Delete, Rename, and Duplicate. Below the table, there are various panels including 'Queries [6]', 'Table.SelectColumns(Source, {City, State, Country, Region})', and 'Column statistics' and 'Value distribution' charts.

Table Data (Visible Rows):

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name
24599	IN-2017-CA120551-428...	22-03-2017	29-03-2017	Standard Class	CA-120551	Cathy Arm
29465	ID-2015-BD116051-422...	01-09-2015	04-09-2015	Second Class	BD-116051	Brian Dah
24598	IN-2017-CA120551-428...	22-03-2017	29-03-2017	Standard Class	CA-120551	Cathy Arm
24597	IN-2017-CA120551-428...	22-03-2017	29-03-2017	Standard Class	CA-120551	Cathy Arm
29464	ID-2015-BD116051-422...	01-09-2015	04-09-2015	Second Class	BD-116051	Brian Dah
28879	ID-2015-AJ107801-421...	19-04-2015	22-04-2015	First Class	AJ-107801	Anthony J
27993	IN-2017-GM144551-429...	01-08-2017	05-08-2017	Standard Class	GM-144551	Gary Mitc
28967	IN-2017-VB217451-430...	11-12-2017	15-12-2017	Standard Class	VB-217451	Victoria
29492	IN-2016-LO171701-426...	24-09-2016	28-09-2016	Standard Class	LO-171701	Lori Olso
28966	IN-2017-VB217451-430...	11-12-2017	15-12-2017	Standard Class	VB-217451	Victoria
25232	ID-2015-SS201401-423...	16-12-2015	20-12-2015	Standard Class	SS-201401	Saphhira
23222	IN-2017-AA103751-429...	10-07-2017	15-07-2017	Second Class	AA-103751	Allen Arm
29094	IN-2015-BG110351-422...	28-09-2015	04-10-2015	Standard Class	BG-110351	Barry Gon
28265	IN-2016-AH105851-427...	27-11-2016	01-12-2016	Standard Class	AH-105851	Angele Ho
27278	IN-2016-CS118451-423...	18-01-2016	20-01-2016	First Class	CS-118451	Cari Sayr
27279	IN-2016-CS118451-423...	18-01-2016	20-01-2016	First Class	CS-118451	Cari Sayr
29096	IN-2015-BG110351-422...	28-09-2015	04-10-2015	Standard Class	BG-110351	Barry Gon
23219	IN-2017-AA103751-429...	10-07-2017	15-07-2017	Second Class	AA-103751	Allen Arm
28276	IN-2014-AH105851-419...	30-11-2014	03-12-2014	First Class	AH-105851	Angele Ho
29585	IN-2015-DW131951-421...	05-06-2015	10-06-2015	Standard Class	DW-131951	David Wie
23951	IN-2014-RS194201-418...	16-08-2014	18-08-2014	First Class	RS-194201	Ricardo S

Queries [6]

Table.SelectColumns(Source, {City, State, Country, Region})

Column statistics

Column	Count
Count	1000
Error	0
Empty	0
Distinct	7
Unique	0
Empty string	0

Value distribution

Region	Count
Southern Asia	7
Oceania	0
North Africa	0
Central Africa	0
Southern Asia	0
Southern Europe	0
Western Asia	0

	A ^B _C City	A ^B _C State	A ^B _C Country	A ^B _C Region
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
	23 distinct, 23 unique	23 distinct, 23 unique	20 distinct, 19 unique	23 distinct, 23 unique
1	Herat	Hirat	Afghanistan	Southern Asia
2	Durres	Durrës	Albania	Southern Europe
3	Algiers	Alger	Algeria	North Africa
4	Benguela	Benguela	Angola	Central Africa
5	Buenos Aires	Buenos Aires	Argentina	South America
6	Vanadzor	Lori	Armenia	Western Asia
7	Canberra	Australian Capital T...	Australia	Oceania

Column statistics		Value distribution	
Count	23	Southern Asia	
Error	0	Southern Europe	
Empty	0	North Africa	
Distinct	23	Central Africa	
Unique	23	South America	
Empty string	0	Western Asia	
Min	Canada	Oceania	
Max	Western...	Western Europe	
		Caribbean	
		Eastern Europe	
		Central America	

Close & Apply | New Source | Recent Sources | Enter Data | Data source settings | Manage Parameters | Export query results | Refresh Preview | Manage Query

Close | New Query | Data Sources | Parameters | Output Data | Query

Queries [7]

- Orders
- People
- Returns
- Dim_Customer
- Dim_Product
- Dim_Region
- Dim_Date**

fx = Table.RenameColumns(#"Removed")

	Date
	Valid 100%
	Error 0%
	Empty 0%
	394 distinct, 152 unique
1	22-03-2017
2	01-09-2015
3	22-03-2017
4	22-03-2017
5	01-09-2015
6	19-04-2015
7	01-08-2017
8	11-12-2017
9	24-09-2016
10	11-12-2017

Enter data

Dim

City

Country

Region

Status

Fact

Customer

Date

Discount

Order

Product

Profit

Quantity

Region

Category

+ New relationship

Autodetect

Edit

Delete

Filter

☐ From: table (column) ↑	Relationship	To: table (column)	Status
☐ Fact_Sales (Customer ID)	* — 1	Dim_Customer (Customer ID)	Active ...
☐ Fact_Sales (Date)	* — 1	Dim_Date (Date)	Active ...
☐ Fact_Sales (Product ID)	* — 1	Dim_Product (Product ID)	Active ...
☐ Fact_Sales (Region)	* — 1	Dim_Region (Region)	Active ...

Paste

Copy

Format painter

Clipboard

Get data

Excel workbook

OneLake catalog

SQL Server

Data

Model view

DAX

TMDL

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p active

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n both

changes

p editor

Copy, Format painter, Get data, Excel workbook, OneLake catalog, SQL Server, Enter data, Data source, Recent sources, Transform data, Refresh data, New visual, Text box, More visuals, New visual calculation, New measure, Quick measure, Sensitivity, Publish, Prep data for Copilot AI, There are pending changes in your queries that haven't been applied. Apply changes, Discard changes, Visualizations, Data

Build visual, Filters, Values, Add data fields here, Drill through, Cross-report, Keep all filters, Add drill-through fields here

Load

- Dim_Customer Loading data...
- Dim_Product Loading data...
- Dim_Region Loading data...
- Dim_Date Loading data...
- Fact_Sales Loading data...

Cancel

File, Home, Transform, Add Column, View, Tools, Help

Close & Apply, New Source, Recent Sources, Enter Data, Data source settings, Manage Parameters, Export query results, Refresh Preview, Properties, Advanced Editor, Choose Columns, Remove Columns, Keep Rows, Remove Rows, Split Column, Group By, Data Type: Date, Use First Row as Headers, Replace Values, Merge Queries, Append Queries, Combine Queries

Queries [8]

- Orders
- People
- Returns
- Dim_Customer
- Dim_Product
- Dim_Region
- Dim_Date
- Fact_Sales

Table.TransformColumnTypes(#"Renamed Columns", {"Date", type date}, "en-GB")

	Order ID	Date	Customer ID	Region	Product ID	Sales	Quantity
1	IN-2017-CA120551-428...	22-03-2017	CA-120551	Southern Asia	FUR-BO-4861	731.82	
2	ID-2015-BD116051-422...	01-09-2015	BD-116051	Southern Asia	OFF-SU-2988	243.54	
3	IN-2017-CA120551-428...	22-03-2017	CA-120551	Southern Asia	TEC-MA-4211	346.32	
4	IN-2017-CA120551-428...	22-03-2017	CA-120551	Southern Asia	FUR-FU-5726	169.68	
5	ID-2015-BD116051-422...	01-09-2015	BD-116051	Southern Asia	OFF-EH-3664	203.88	
6	ID-2015-AJ107801-421...	19-04-2015	AJ-107801	Southern Asia	FUR-TA-3420	4626.15	
7	IN-2017-GM144551-429...	01-08-2017	GM-144551	Southern Asia	FUR-BO-3626	2070.15	
8	IN-2017-VB217451-430...	11-12-2017	VB-217451	Southern Asia	FUR-CH-4683	914.34	
9	IN-2016-LO171701-426...	24-09-2016	LO-171701	Southern Asia	TEC-PH-5814	1168.44	
10	IN-2017-VB217451-430...	11-12-2017	VB-217451	Southern Asia	TEC-PH-3129	500.94	
11	ID-2015-SS201401-423...	16-12-2015	SS-201401	Southern Asia	FUR-TA-3780	1447.56	
12	IN-2017-AA103751-429...	10-07-2017	AA-103751	Southern Asia	OFF-AP-4747	669.12	
13	IN-2015-BG110351-422...	28-09-2015	BG-110351	Southern Asia	TEC-CO-4588	427.41	
14	IN-2016-AH105851-427...	27-11-2016	AH-105851	Southern Asia	FUR-CH-4523	417.42	
15	IN-2016-CS118451-423...	18-01-2016	CS-118451	Southern Asia	OFF-ST-6068	332.85	
16	IN-2016-CS118451-423...	18-01-2016	CS-118451	Southern Asia	FUR-TA-3758	267.09	
17	IN-2015-BG110351-422...	28-09-2015	BG-110351	Southern Asia	TEC-PH-5251	412.95	
18	IN-2017-AA103751-429...	10-07-2017	AA-103751	Southern Asia	OFF-ST-5708	244.8	

Query Settings

PROPERTIES

- Name: Fact_Sales
- All Properties

APPLIED STEPS

- Source
- Removed Ot
- Renamed Cc
- Changed Ty

Power BI Desktop interface showing a query editor with a table named "Table.RenameColumns(...)" and a "Change Type with Locale" dialog box.

The table columns are: Order ID, Date, Customer ID, Region, Product ID, Sales, and Quantity. The "Date" column is highlighted, and the dialog box is open, showing the "Data Type" set to "Date" and the "Locale" set to "English (United Kingdom)".

The "APPLIED STEPS" pane on the right shows the following steps:

- Source
- Removed Other Columns
- Renamed Columns

The "PROPERTIES" pane on the right shows the "Name" property set to "Fact_Sales".

The "Query Settings" pane on the right shows the "Name" property set to "Dim_Date".

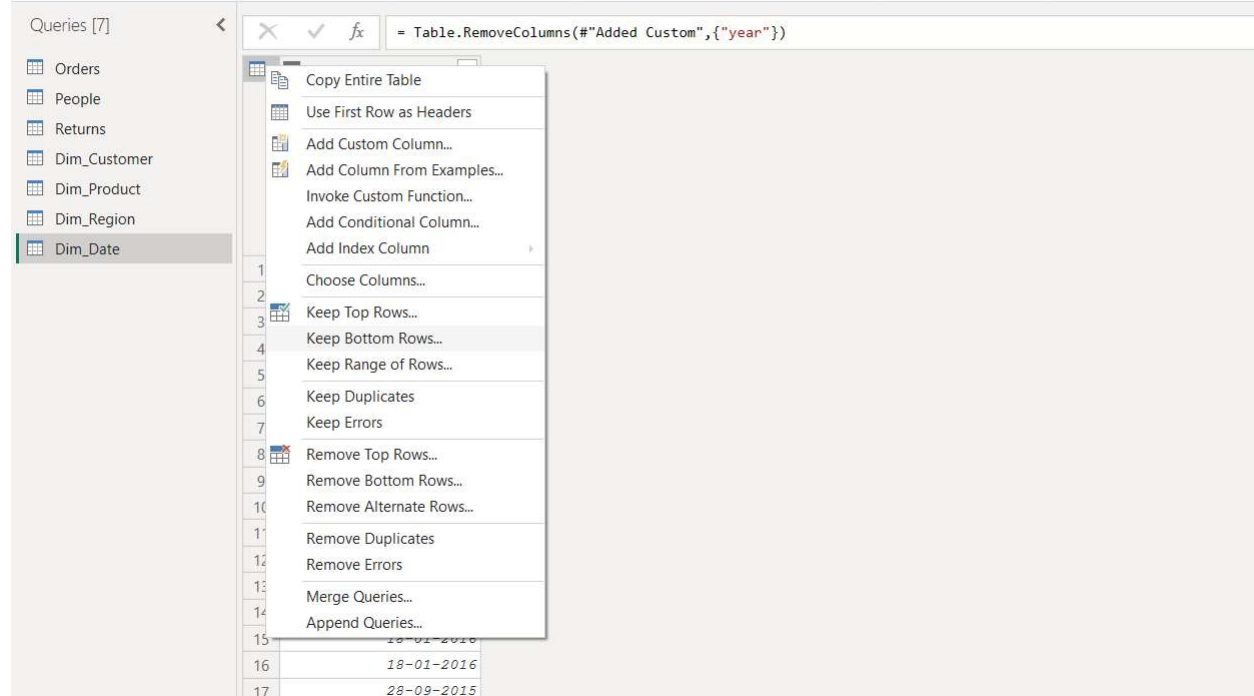
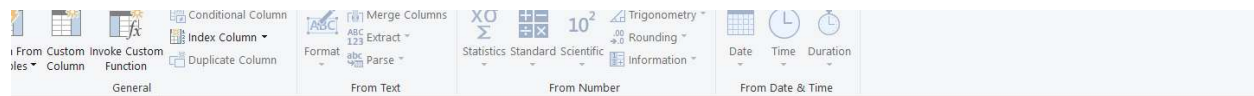
The "APPLIED STEPS" pane on the right shows the following steps:

- Source
- Removed Other Columns
- Renamed Column
- Filtered Rows
- Changed Type with Locale
- Added Custom Column
- Removed Column
- Duplicated Column
- Removed Column
- Inserted Year
- Inserted Quarter
- Inserted Month
- Inserted Month Filter
- Removed Column
- Removed Duplicate

The "Table.Distinct(...)" query is shown below the "Table.RenameColumns(...)" query. The table columns are: Date, Year, Quarter, and Month Name. The "Date" column is highlighted, and the dialog box is open, showing the "Data Type" set to "Date" and the "Locale" set to "English (United Kingdom)".

The "APPLIED STEPS" pane on the right shows the following steps:

- Source
- Removed Other Columns
- Renamed Column
- Filtered Rows
- Changed Type with Locale
- Added Custom Column
- Removed Column
- Duplicated Column
- Removed Column
- Inserted Year
- Inserted Quarter
- Inserted Month
- Inserted Month Filter
- Removed Column
- Removed Duplicate



ie Data Type: Date 1 2 Replace Values U English (Türks and Caicos Islands)

Rows Detect Data Type Fill M English (Tuvalu)

WS Rename Pivot Column C English (Uganda)

Any Column English (United Arab Emirates)

English (United Kingdom)

English (United States)

English (US Minor Outlying Islands)

English (US Virgin Islands)

English (Vanuatu)

English (World)

English (Zambia)

English (Zimbabwe)

Esperanto (World)

Estonian (Estonia)

Ewe (Ghana)

Ewe (Togo)

Ewondo (Cameroon)

Faroese (Denmark)

Faroese (Faroe Islands)

Filipino (Philippines)

English (India)

394 distinct, 152 unique

235 20-08-2017

236 25-06-2014

237 07-11-2017

238 14-08-2017

239 27-12-2017

240 20-05-2017

241 28-05-2015

242 09-06-2017

243 21-12-2014

244 09-06-2017

245 09-06-2017

246 31-08-2015

247 09-06-2017

248 09-06-2017

249 31-08-2015

250 09-06-2017

Sample input values:

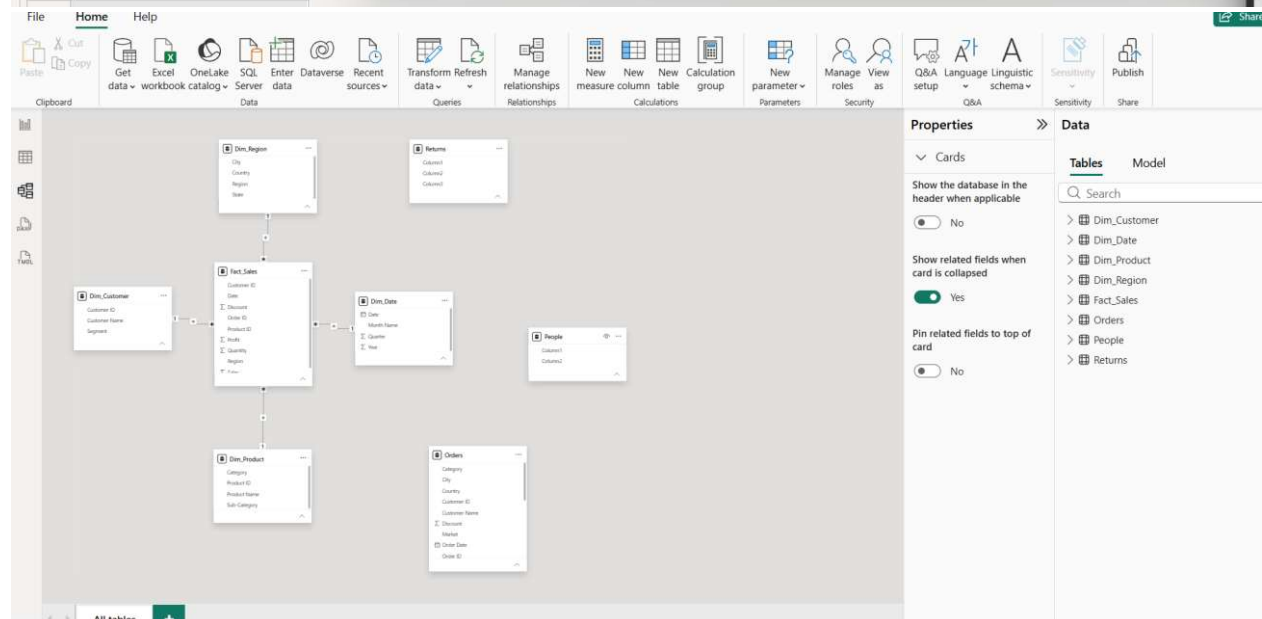
29-03-2016

29 March 2016

29 March

March, 2016

OK





 DAX queries will be saved to your model. They won't be visible when published in the Power BI service. [Learn more](#) 



 Run

 Update model with changes (0)



1 

 write the dax queries with correct query  

Copilot uses AI. Always review content for mistakes. [Read terms](#)



```
2 Total Profit = SUM(Fact_Sales[Profit])
3 Profit Margin = DIVIDE( SUM(Fact_Sales[Profit]), SUM(Fact_Sales[Sales]) )
4 Average Order Value = AVERAGE(Fact_Sales[Sales])
5 Total Quantity = SUM(Fact_Sales[Quantity]).
```

Results